02 BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP

File: group_sheets/02_bus_select.kicad_sch

03 SUBGROUP

File: group_sheets/03_subgroup.kicad_sch

04 SUBGROUP, AUX MASTER

File: group_sheets/04_subgroup.kicad_sch

05 AUX MASTER, STEREO RETURN

File: group_sheets/05_aux_master.kicad_sch

06 STEREO RETURN

File: group_sheets/06_stereo_return.kicad_sch

07 STEREO RETURN

File: group_sheets/07_stereo_return.kicad_sch

08 JACK BOARD LINK, POWER AND GROUNDS

File: group_sheets/08_jack_board_link.kicad_sch

09 RIBBON

File: group_sheets/09_ribbon.kicad_sch

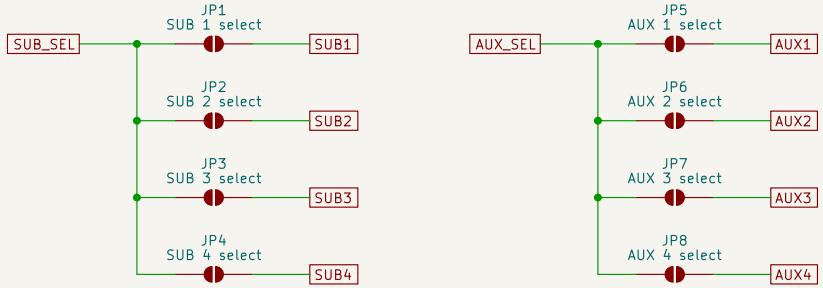
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)

Sheet: /
File: group.kicad_sch

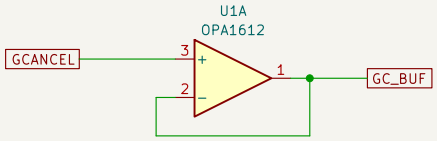
Title: Group module — goa console (sub + return + aux master)

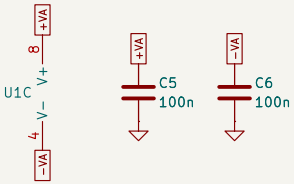
Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 1/9

BUS SELECT close ONE SUB jumper and ONE AUX jumper: group n sums SUB n and AUX n (all open = summer idle)

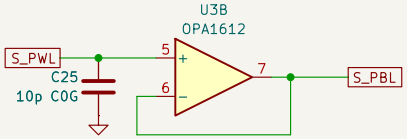


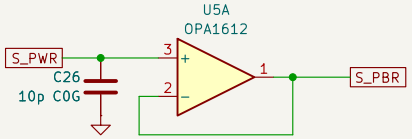
GROUND CANCEL BUFFER (local, for this module's re-inverters; summing amps use the raw GCANCEL line)

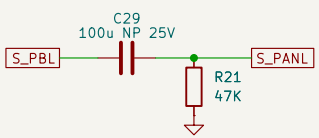


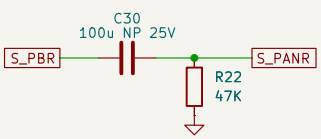


SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL [spec 12]









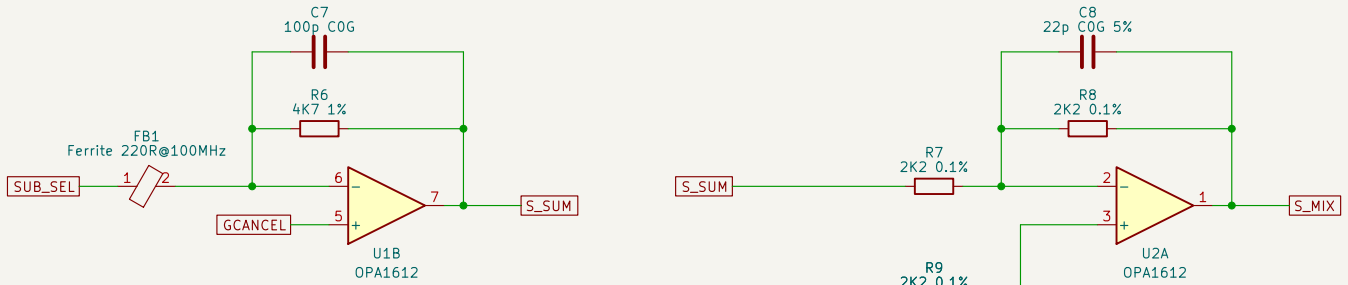
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

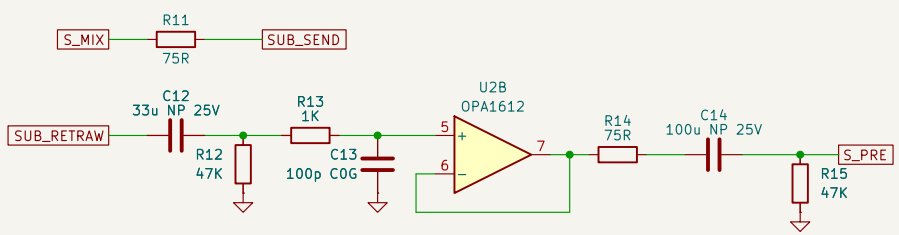
Sheet: /02 BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP/
File: 02_bus_select.kicad_sch

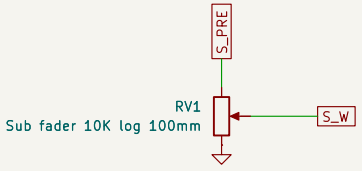
Title: BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP

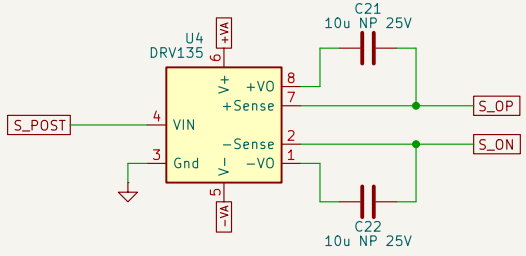
Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 2/9

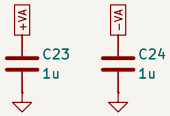
SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL (cont.)

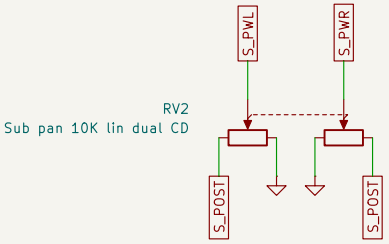


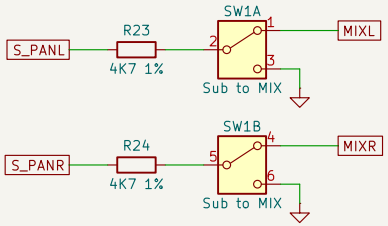












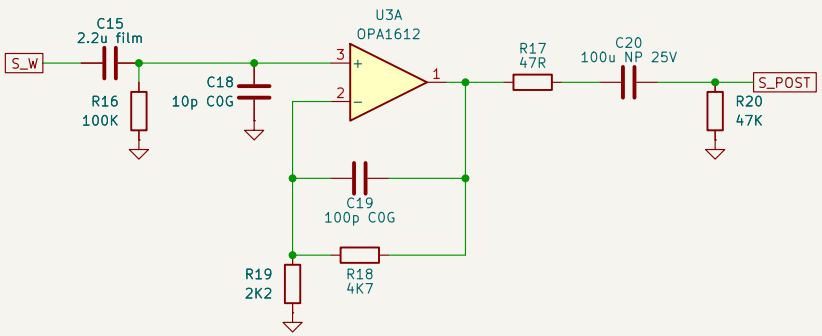
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /03 SUBGROUP/
File: 03_subgroup.kicad_sch

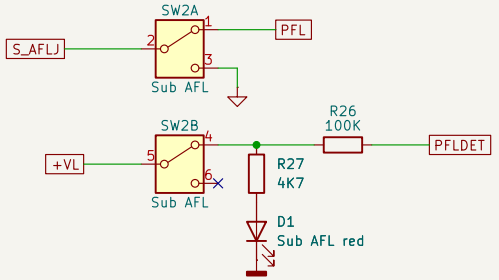
Title: SUBGROUP

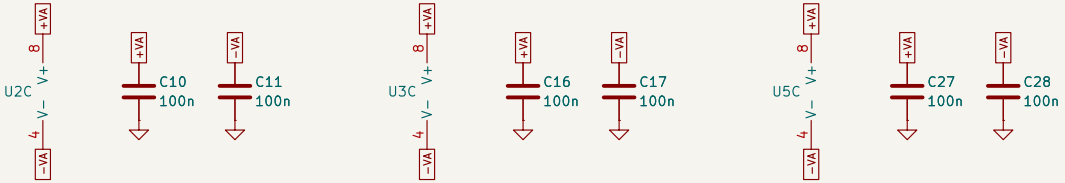
Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 3/9

SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL (cont.)

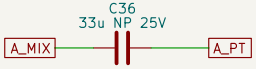


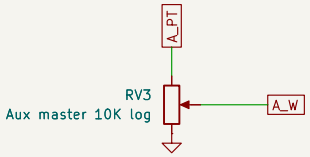


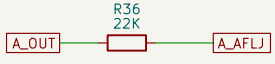


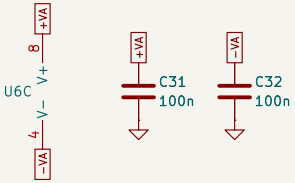


AUX MASTER: OPA1612 sum (selected AUX bus, 22K || 100p) + re-invert, level, buffer, DRV134 out, AFL [spec 13]









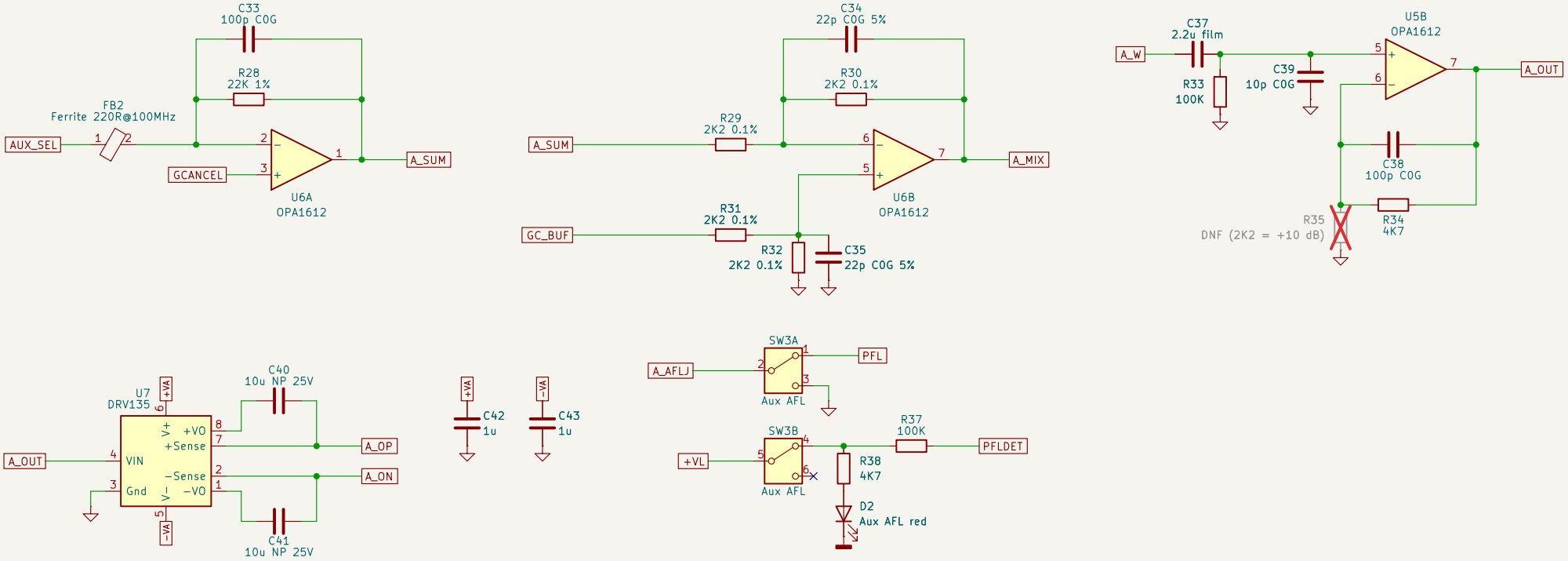
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /04 SUBGROUP, AUX MASTER/
File: 04_subgroup.kicad_sch

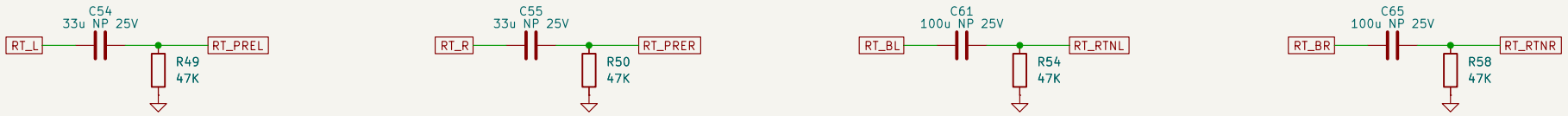
Title: SUBGROUP, AUX MASTER

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 4/9

AUX MASTER: OPA1612 sum (selected AUX bus, 22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)



STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1-2 / 3-4 onto the ribbon, PFL [spec 16]



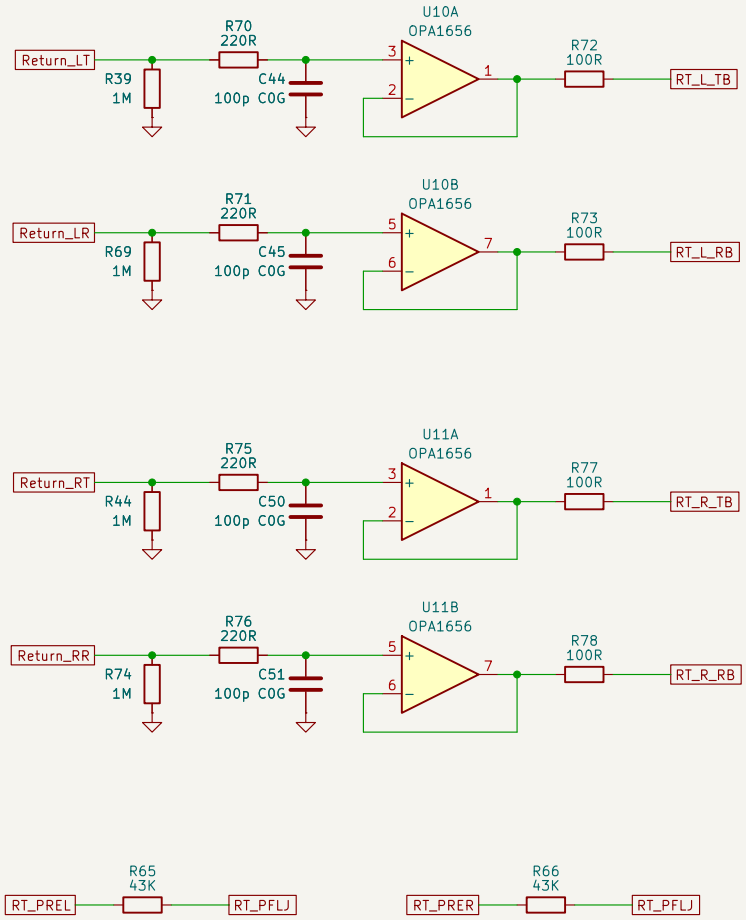
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

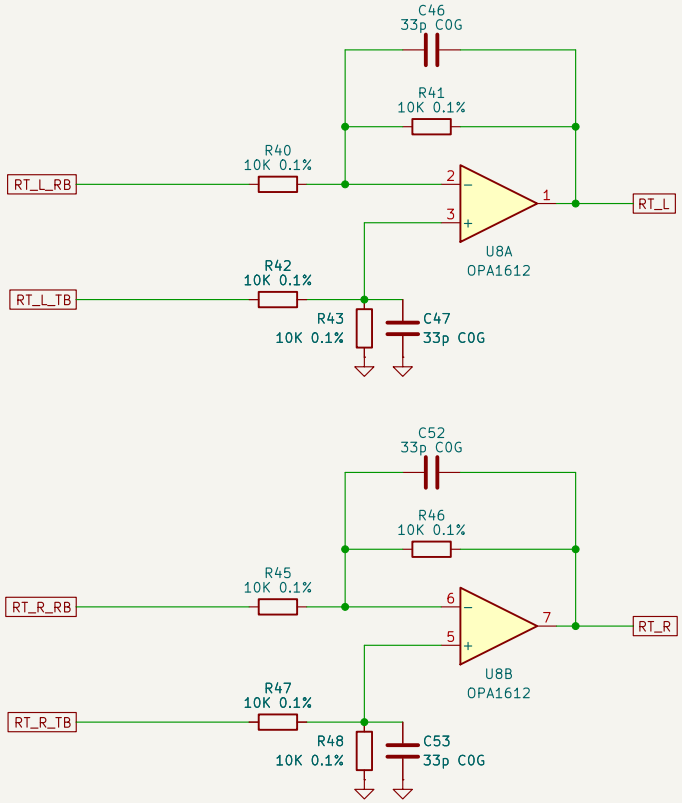
Sheet: /05 AUX MASTER, STEREO RETURN/
File: 05_aux_master.kicad_sch

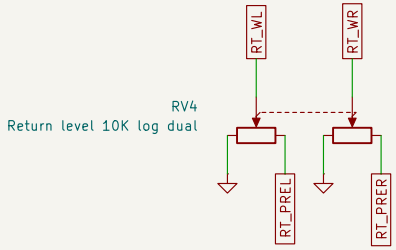
Title: AUX MASTER, STEREO RETURN

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 5/9

STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1–2 / 3–4 onto the ribbon, PFL (cont.)







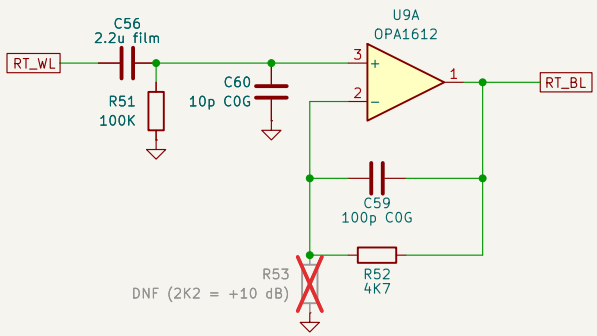
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

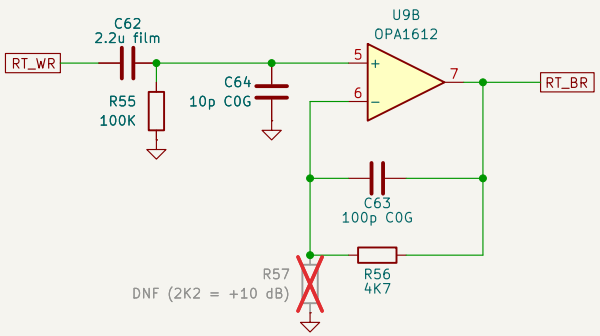
Sheet: /06 STEREO RETURN/
File: 06_stereo_return.kicad_sch

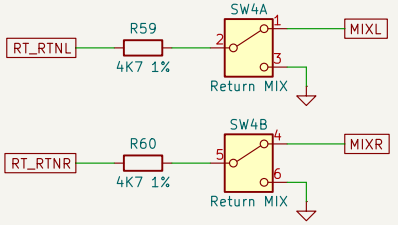
Title: STEREO RETURN

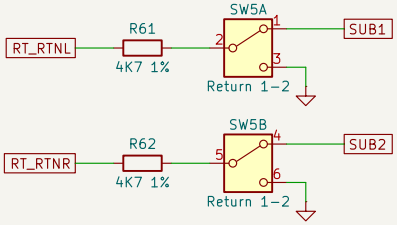
Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 6/9

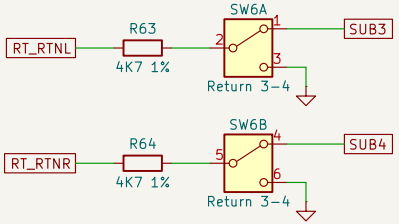
STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1–2 / 3–4 onto the ribbon, PFL (cont.)

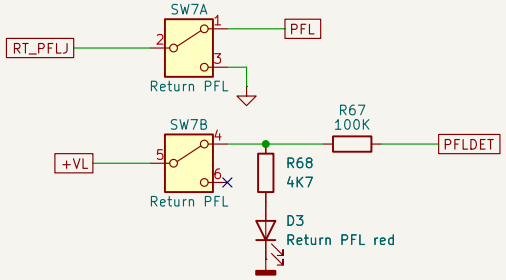


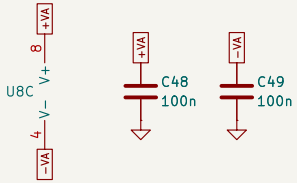


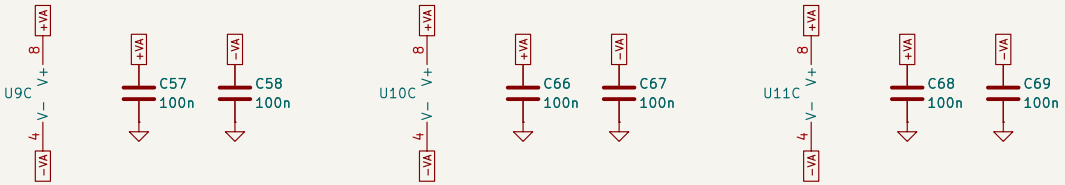












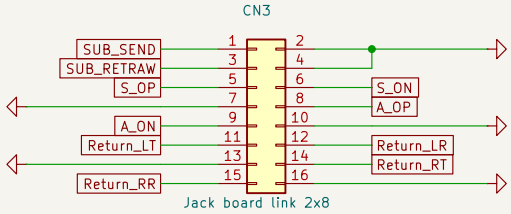
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /07 STEREO RETURN/
File: 07_stereo_return.kicad_sch

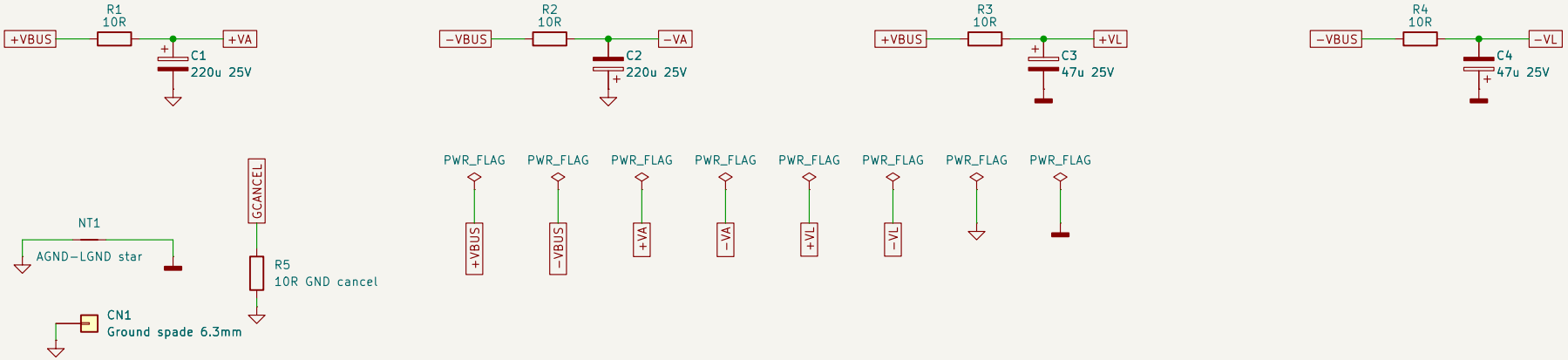
Title: STEREO RETURN

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 7/9

JACK BOARD LINK 2x8 IDC ribbon to the group jack board (5 jacks); bottom side, as the strip's CN3



POWER AND GROUNDS (as a strip: 10R + 220u per audio rail, 10R + 47u logic rails, spade, 10R ground cancel)



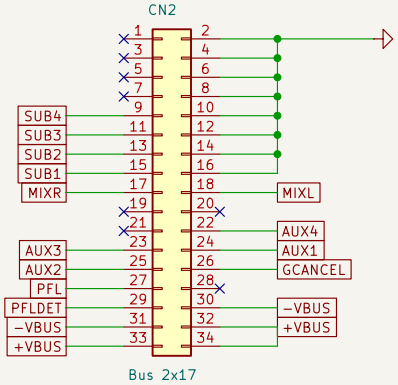
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /08 JACK BOARD LINK, POWER AND GROUNDS/
File: 08_jack_board_link.kicad_sch

Title: JACK BOARD LINK, POWER AND GROUNDS

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 8/9

RIBBON one 34-pin header, Delta pinout (as a strip); AUX 5-6 are summed on the master module



Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /09 RIBBON/
File: 09_ribbon.kicad_sch

Title: RIBBON

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 9/9